Personality & Emotion for AI Design

BCSAI - Spring 2025

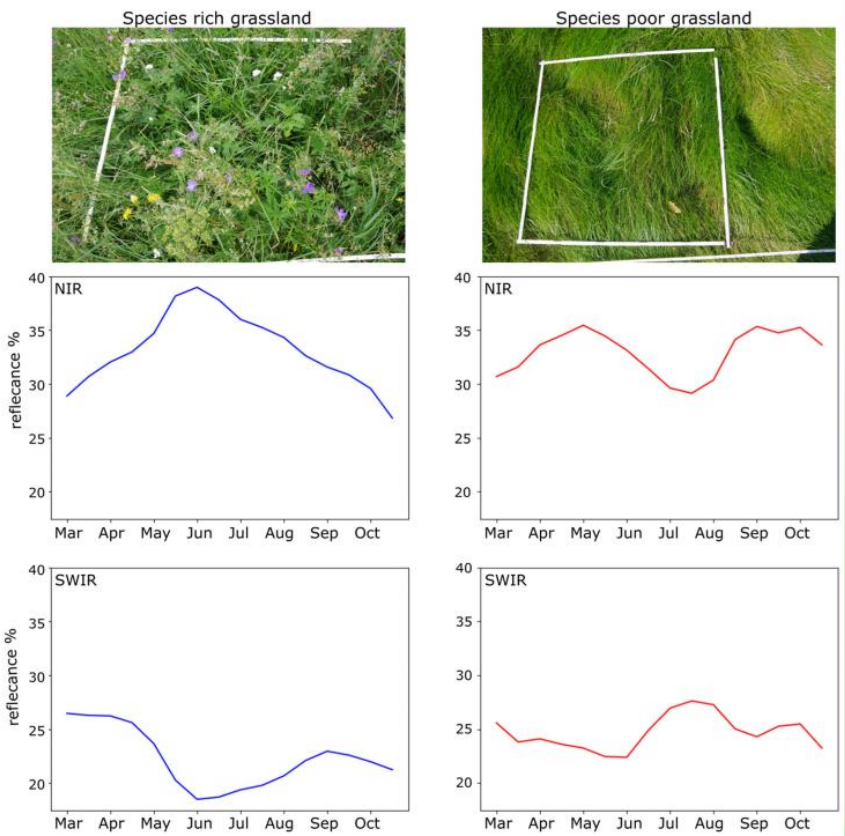
Lecture 1 - Introduction

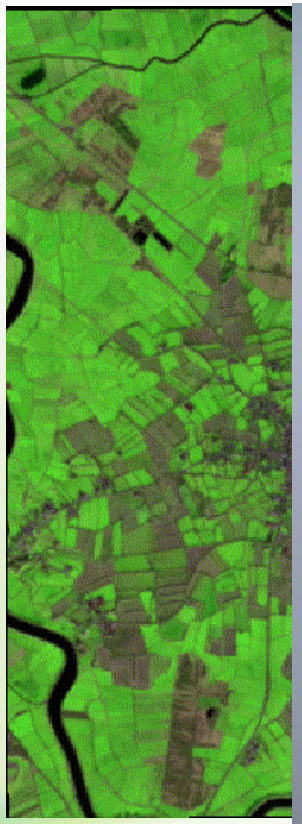
Many slides adapted from Lafferty (Yale), Weinberger (Cornell), Szameitat (Brunel)

Me

Javier Muro

- Remote sensing scientists
- Geospatial data analysis and applied AI





Course objectives

- Gain some new perspective
- Appreciate some of the **power** and **limitations** of ML
- Reflect on societal implications of AI/ML
- Learn the basics of **CogSci**, **ML**, **Deep Learning**
- Have fun & want to learn more

Learning objectives

- AI -> major part of your degree
- Many of you will be involved in designing or implementing these systems in the near future

- We should all be able to talk about/have an informed opinion on:
  - the possibilities of “AI”
  - its shortcomings
  - what it is and what it is not

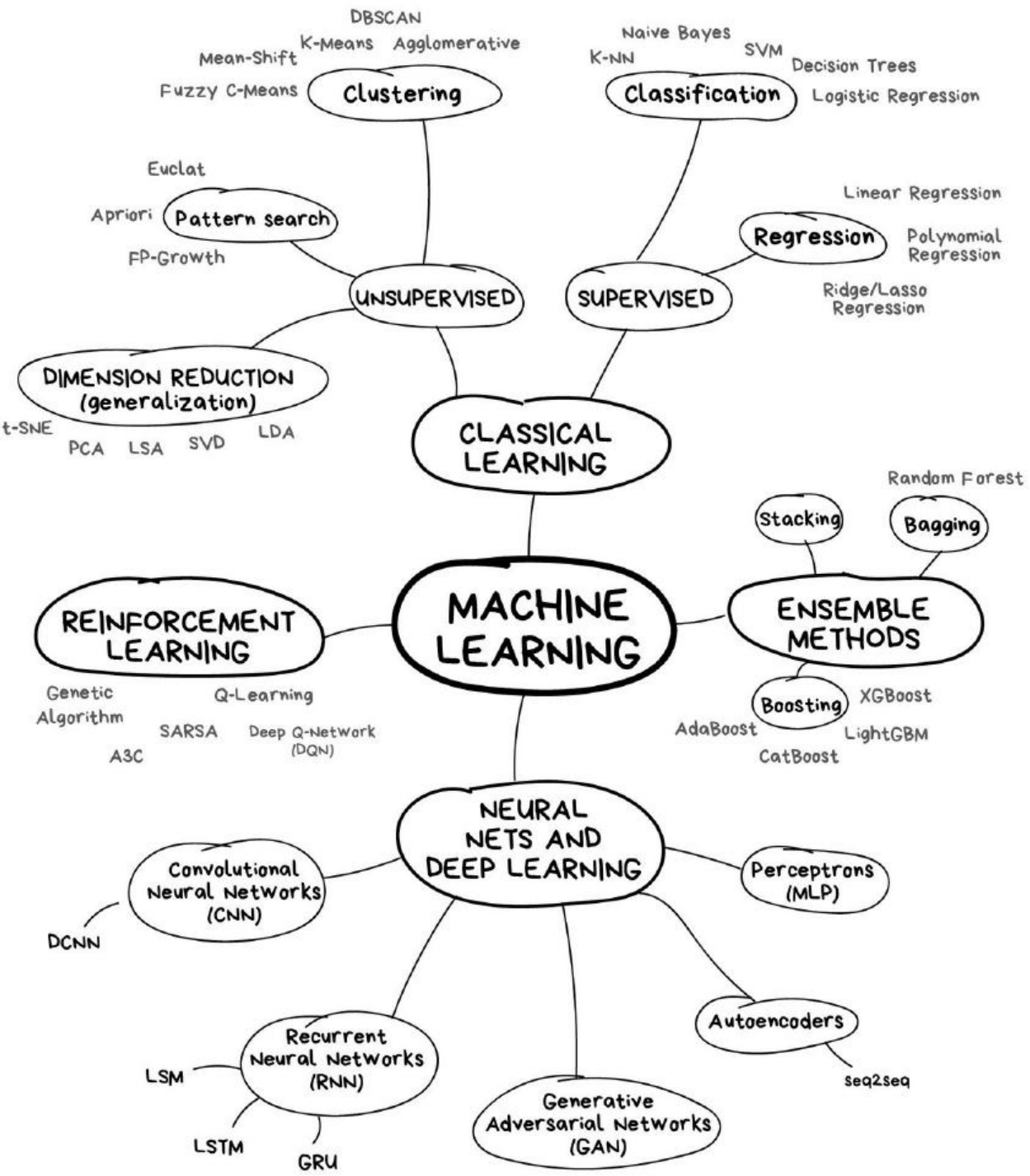
Evaluation criteria

Criteria	Score %
Final Exam	40 %
Group Presentation	30 %
Individual work	20 %
Class Participation	10 %

Ingredients for good lectures

- Professor

- Students



What the current media refers to when use the term "AI" is only a very small part of everything that "intelligent" computer programs can do!

Where everything started

August 31, 1955

John McCarthy, Marvin L. Minsky, Nathaniel Rochester,
and Claude E. Shannon

A Proposal for the

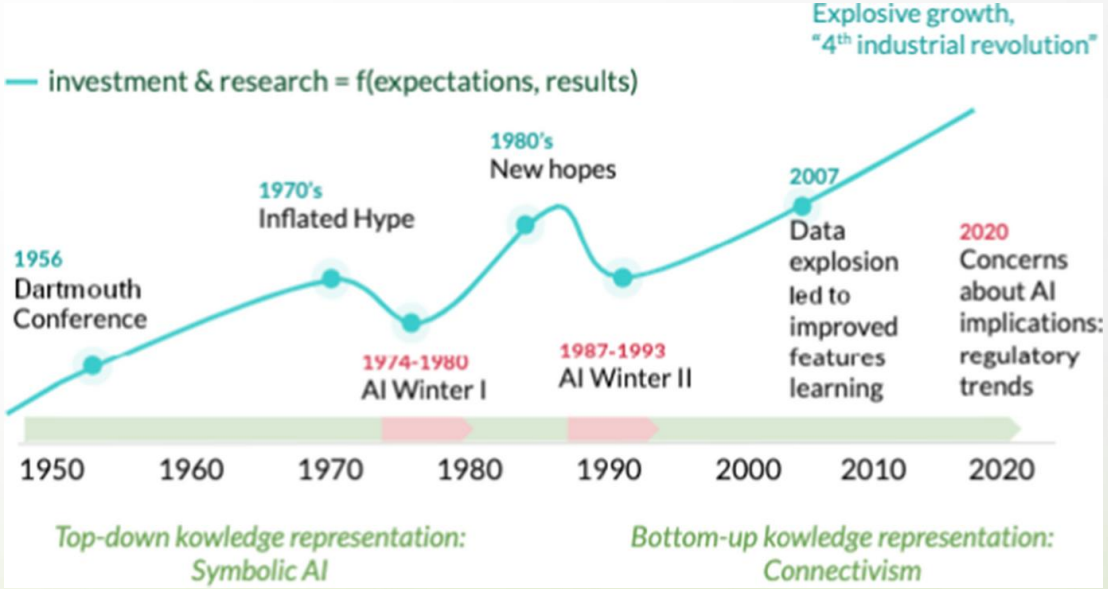
DARTMOUTH SUMMER RESEARCH PROJECT ON ARTIFICIAL INTELLIGENCE

June 17 - Aug. 16

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

Promise of the field of AI (since the 50's and today): ***develop machines that possess intelligence comparable to that of humans***

Where everything started



https://techgenies.com/ai-history-the-first-summer-and-winter-of-ai/

Limitations of AI

Limitations of AI

MAN VS. MACHINE

“Superhuman” *Go* AIs still have trouble defending against these simple exploits

Plugging up “worst-case” algorithmic holes is proving more difficult than expected.

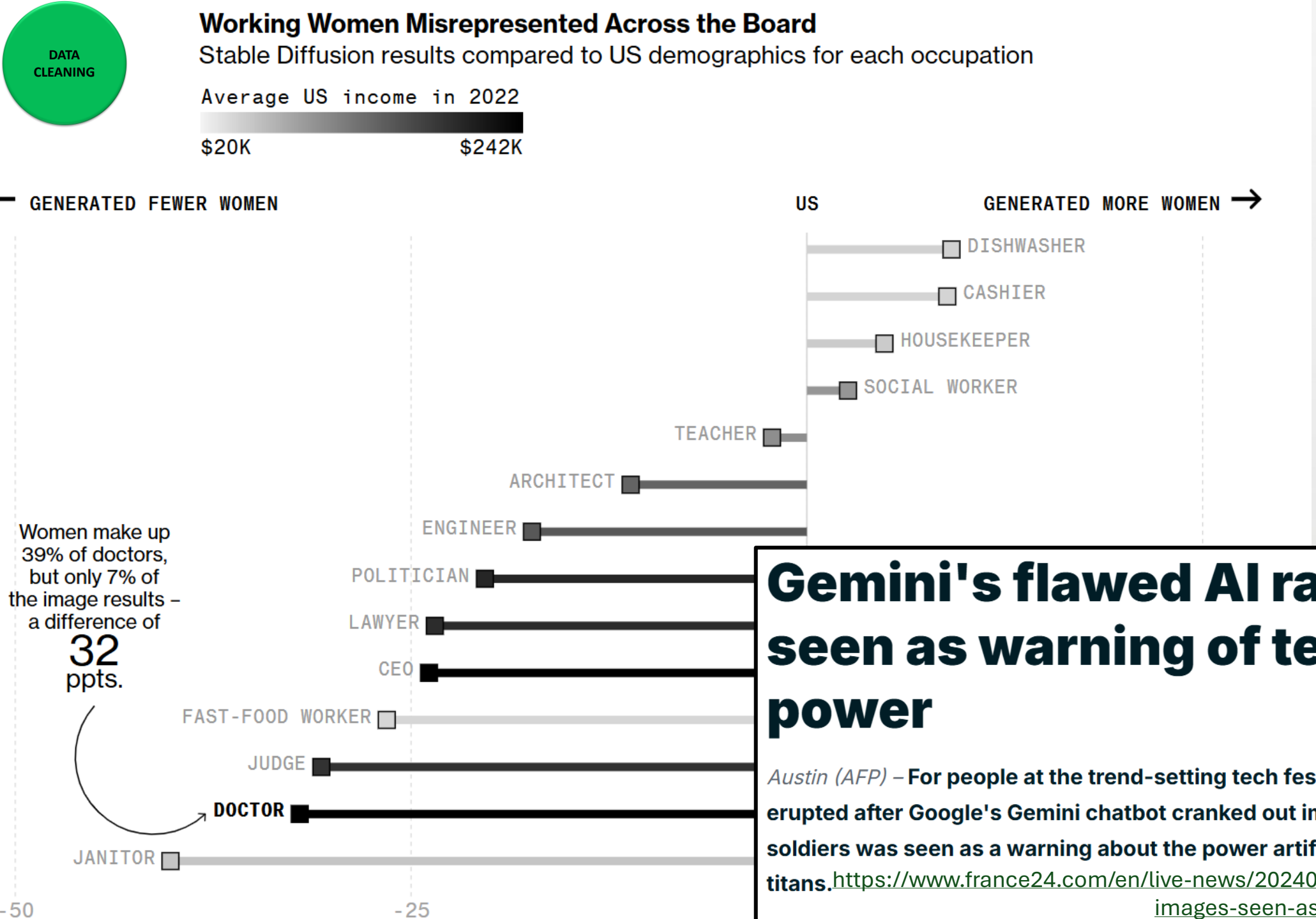
KYLE ORLAND – JUL 12, 2024 8:20 PM | 103

In the ancient Chinese game of *Go*, state-of-the-art artificial intelligence has generally been able to defeat the best human players since at least 2016. But in the last few years, researchers have discovered flaws in these top-level AI *Go* algorithms that give humans a fighting chance. By using unorthodox “cyclic” strategies—ones that even a beginning human player could detect and defeat—a crafty human can often exploit gaps in a top-level AI’s strategy and fool the algorithm into a loss.

DATA CLEANING

Bias in Bias out

https://www.bloomberg.com/graphics/2023-generative-ai-bias/



Gemini's flawed AI racial images seen as warning of tech titans' power

Austin (AFP) – **For people at the trend-setting tech festival here, the scandal that erupted after Google's Gemini chatbot cranked out images of Black and Asian Nazi soldiers was seen as a warning about the power artificial intelligence can give tech titans.** https://www.france24.com/en/live-news/20240317-gemini-s-flawed-ai-racial-images-seen-as-warning-of-tech-titans-power

Join at menti.com | use code 2356 4267

AI will solve all our problems

AI will kill us all

Not much will change

The current AI boom has been spurred by the following principle:

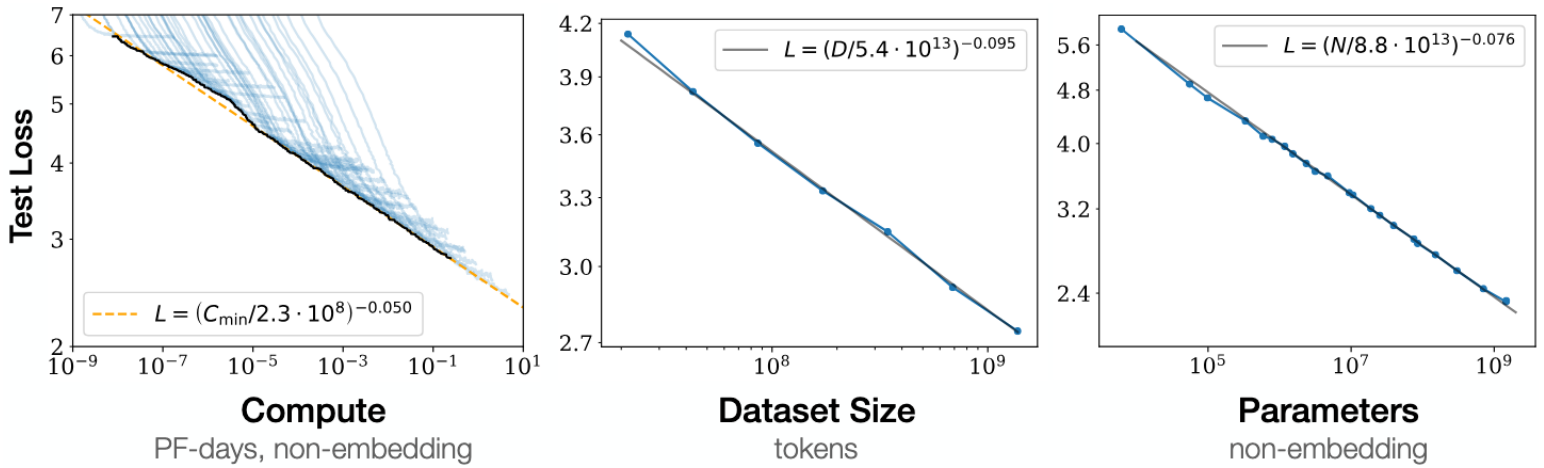


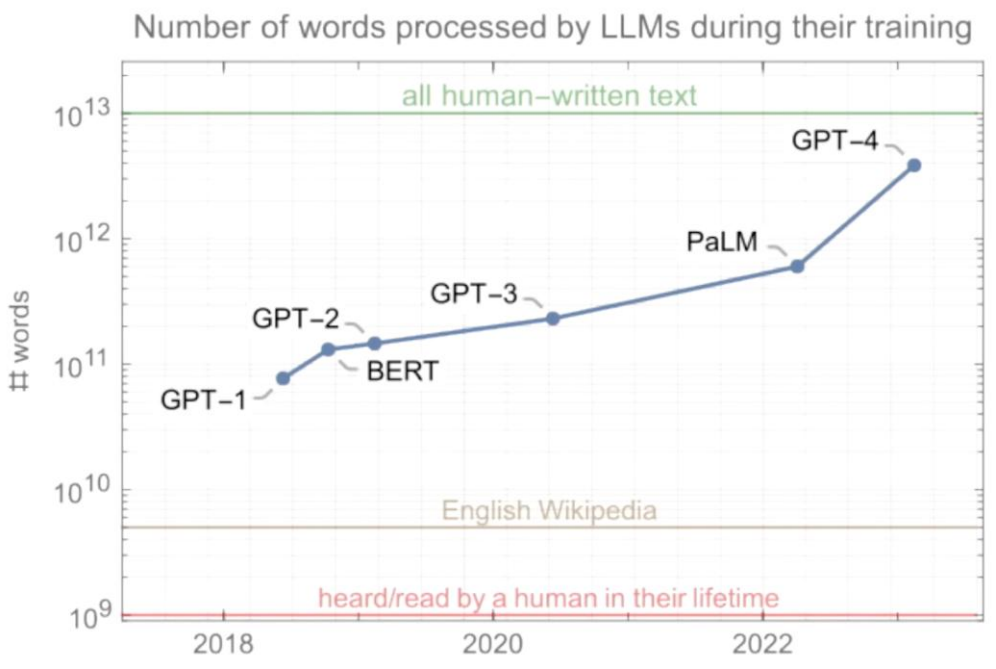
Figure 1 Language modeling performance improves smoothly as we increase the model size, datasetset size, and amount of compute[2] used for training. For optimal performance all three factors must be scaled up in tandem. Empirical performance has a power-law relationship with each individual factor when not

Link to video

The current AI boom has been spurred by the following principle:

massive amounts of data
— often derived from human behavior —
as a *substitute* for intelligence

Now compare with the human brain:



Source: https://www.youtube.com/watch?v=_6R7Ym6Vy_I

Criticism from part of the AI/tech community:

GARY MARCUS

DEC 08, 2025

rob pike
@robpike.io
Follow

F[redacted]k you people. R[redacted]g the planet, spending trillions on toxic, unrecyclable equipment while blowing up society, yet taking the time to have your vile machines thank me for striving for simpler software.

Just f[redacted]k you. F[redacted]k you all.

I can't remember the last time I was this angry.

Richard Sutton – Father of RL thinks LLMs are a dead end

https://www.youtube.com/watch?v=21EYKqUsPfg

Efficiency

A. Energy and the Brain

The average person's resting metabolic rate (RMR) is about 1,300 calories/day. The brain uses approximately 20% of this. Convert the brain's daily energy use into joules, then calculate its power use in watts. **Calculate the human brain's power consumption using the data provided.**

~13 Watts

*(**Hint:** Power is the amount of energy consumed per second, and RMR is the amount of energy consumed per day.)*

B. The Human Brain's Computational Efficiency

The brain is estimated to be capable of performing on average 1×10^{18} FLOPs (floating-point operations per second). **Determine the ratio of FLOPs of computation per watt of brain function.**

2.8×10^{18} FLOPs/Watt

Source: https://craft.stanford.edu/resource/is-ai-more-efficient-than-the-human-brain/

C. Energy and the AI Data Center

Estimates suggest LLM AI data centers use several thousand GPUs to answer users' queries. A conservative estimate of 2,000 GPUs, each requiring 1,000 watts of power, can give a very rough sense of how much energy an AI data center consumes. **Calculate the power consumption of an AI data center to answer people's queries (a.k.a. prompts or inferences), assuming the GPUs are using 70% of their full energy potential.**

~1 400 000 Watts

(Hint: Remember, power is the amount of energy used per second.)

D. AI Data Center's Computational Efficiency

Let's assume an AI data center is using an NVIDIA HGX 200, with each GPU capable of performing 1.44×10^{17} FLOPs. **Determine the ratio of FLOPs of computation per watt of GPU power consumption.**

2.1×10^{14} FLOPs/Watt

E. Human vs AI Data Center Computational/Energy Efficiency

How many times greater is the ratio of FLOPs/watt of the human brain compared to an AI data center?

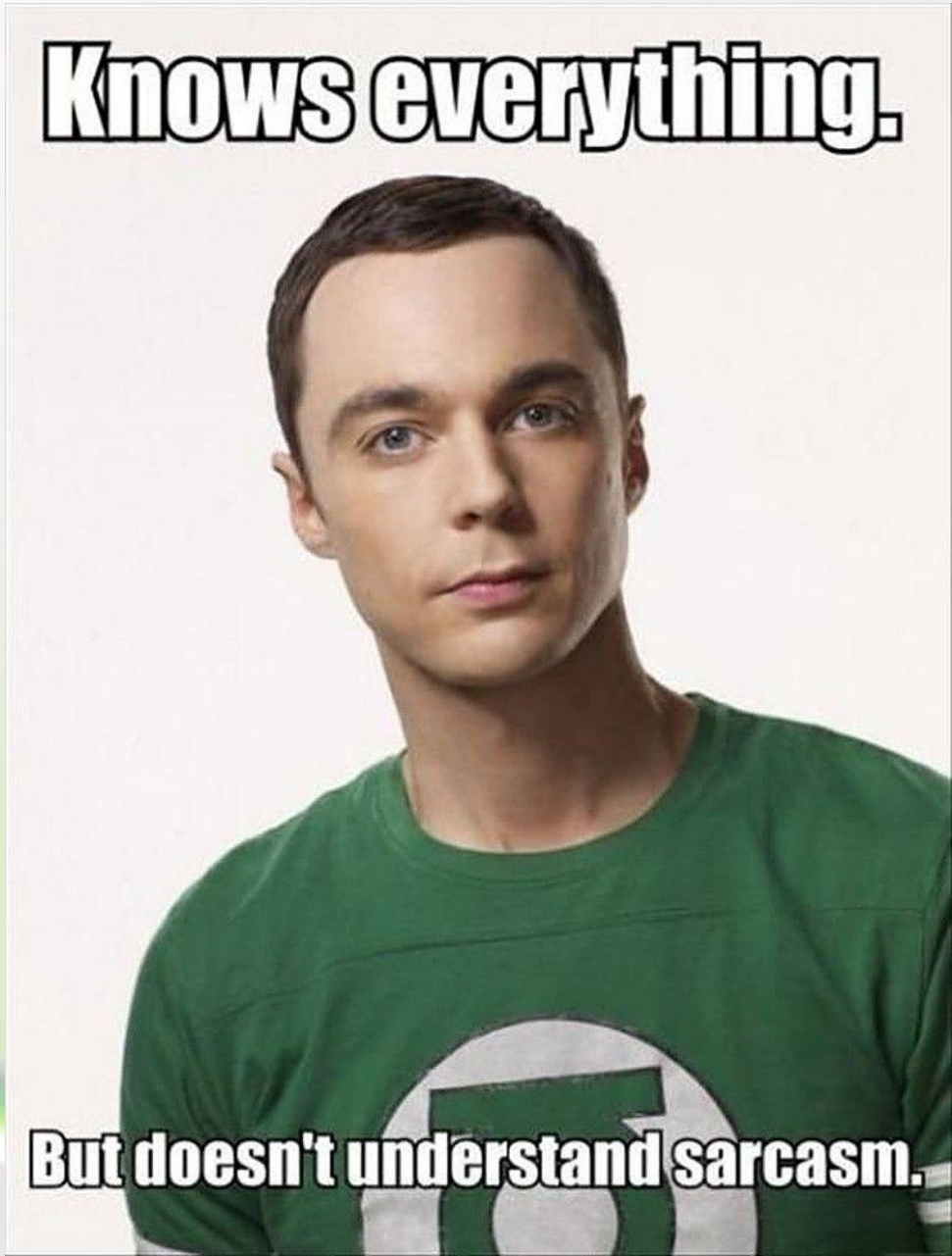
The human brain is approx. 13,446 times more energy efficient!

Source: https://craft.stanford.edu/resource/is-ai-more-efficient-than-the-human-brain/

Defining intelligence: two models

One general intelligence: people with high intelligence perform well across many disciplines

Multiple intelligences: **not supported empirically** in humans. There are cognitive domains/talents (memory, reasoning, spatial or mathematical skills) where some excell -> "narrow AI"



Intelligence

- Can animals be intelligent truly?

https://www.youtube.com/watch?v=IQW_K8rq1N4

https://www.youtube.com/watch?v=ZHpu7ngQxwE

Defining intelligence

In 2007, Legg and Hutter summarized roughly 70 definitions from the literature into a single statement:

“Intelligence measures an agent’s ability to achieve goals in a wide range of environments.”

“achieving goals”

“in a wide range of environments”

Undwer this view, an **intelligent agent** would achieve high skill across **many different tasks**

Defining intelligence

CRITIC: Then wouldn't you be forced to say that evolution itself must be intelligent, since it solved those problems of flying and building reefs and nests?

No, because people also use the word "intelligence" to emphasize swiftness and efficiency. Evolution's time rate is so slow that we don't see it as intelligent, even though it finally produces wonderful things we ourselves cannot yet make. Anyway, it isn't wise to treat an old, vague word like "intelligence" as though it must define any definite thing. Instead of trying to say what such a word "means," it is better simply to try to explain how we use it.

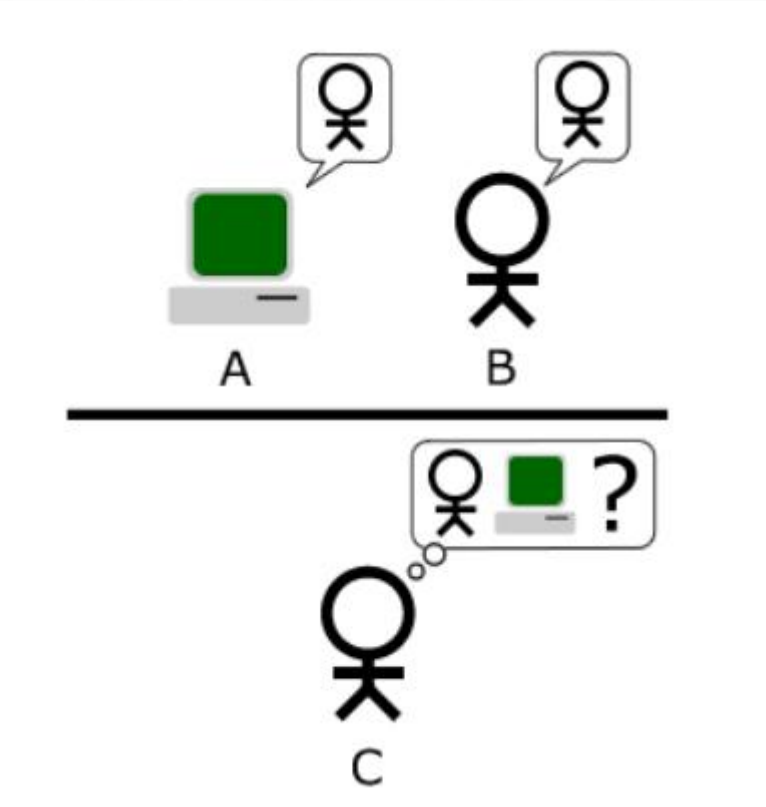
Our minds contain processes that enable us to solve problems we consider difficult. "Intelligence" is our name for whichever of those processes we don't yet understand.

Some people dislike this "definition" because its meaning is doomed to keep changing as we learn more about psychology. But in my view that's exactly how it ought to be, because the very concept of intelligence is like a stage magician's trick. Like the concept of "*the unexplored regions of Africa*," it disappears as soon as we discover it.

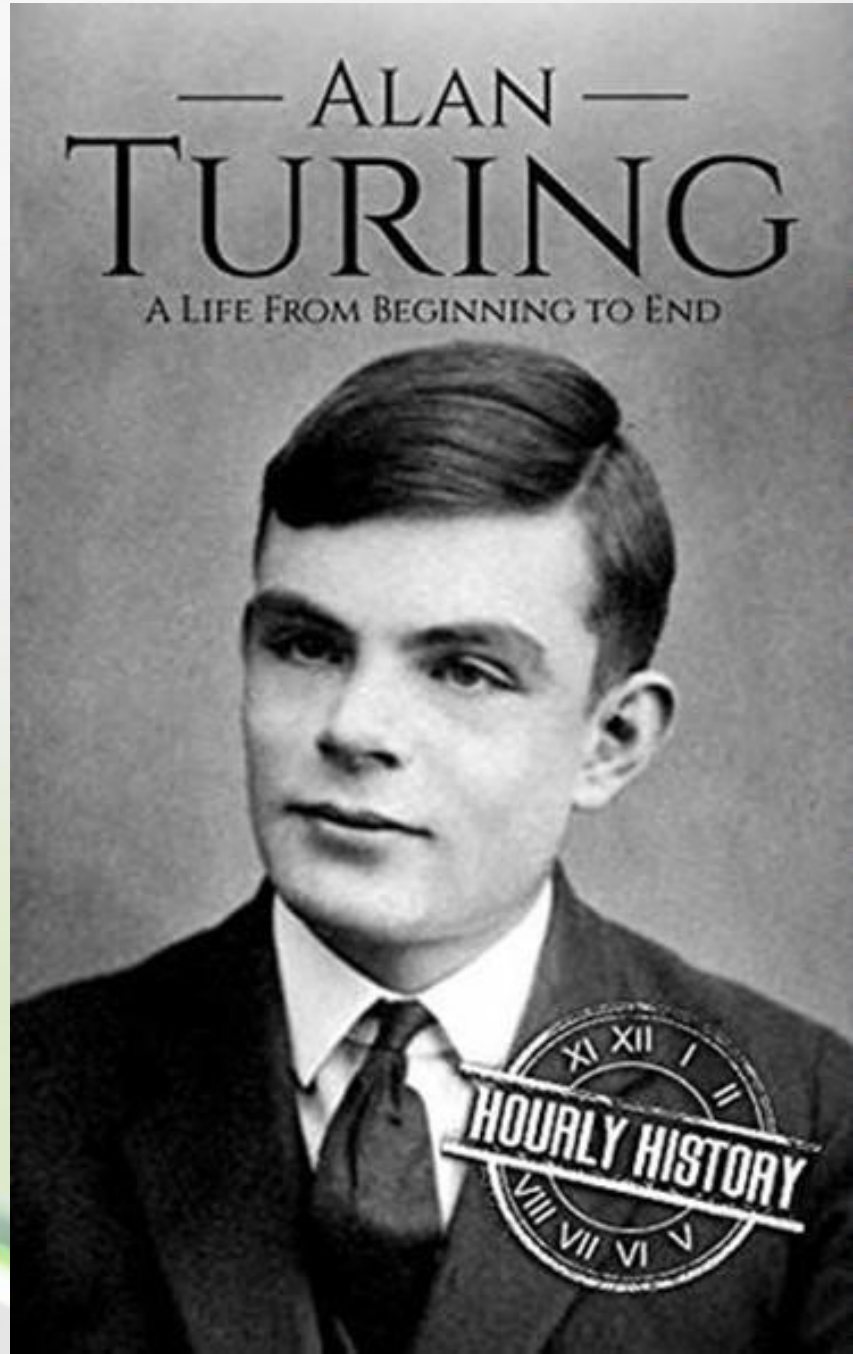
Marvin Minsky, The Society of Mind, 1986

Alan Turing's Imitation Game

Can machines think?



Instead: "Can machines do what we (as thinking entities) can do?"



These used to be considered AI:

- Playing Checkers and Chess (at non-elite levels): Game-playing was the benchmark for intelligence (e.g., early chess programs). Now: Only superhuman performance is considered AI-worthy; basic play is trivial.

Shift point: After Deep Blue (1997), chess engines became “just algorithms.”

- Search and Information Retrieval: Early search engines were seen as exhibiting intelligence—finding relevant information from huge datasets.

Shift point: Web-scale search in the late 1990s–2000s.

- Spell Checking & Grammar Correction: Detecting spelling or grammatical mistakes was viewed as linguistic intelligence.

Shift point: 1980s–1990s word processors.

- Optical Character Recognition (OCR): Teaching a machine to “read” printed text was considered a major AI milestone.

Shift point: 1990s–2000s mass digitization.

These used to be considered AI:

- Face Detection (not recognition): Detecting a human face in an image felt highly intelligent.

Shift point: Early 2000s (e.g., digital cameras with face boxes).

- Speech-to-Text (Basic Dictation): Converting spoken language into text was a core AI challenge.

Shift point: 2010s with reliable mobile dictation.

- Recommendation Systems (Simple Collaborative Filtering): Suggesting movies or products seemed AI-like. Now: Basic personalization; only advanced models draw attention.

Shift point: Late 2000s.

- Industrial Robotics (Fixed Tasks): Robots assembling cars were described as “intelligent machines.”

Shift point: When their lack of adaptability became obvious.

- **Generative AI**

Programs that could generate text and images based on a prompt were considered to be close to AGI.

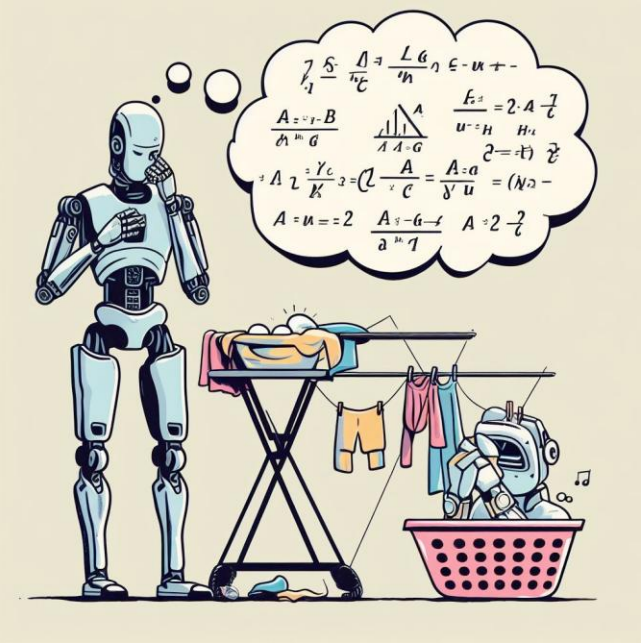
Shift point: When their lack of adaptability **becomes** obvious???

Recap

Although we are able to engineer systems that perform extremely well on specific tasks ("narrow AI"), they have still stark and dangerous limitations.

- ❌ data-hungry / data dependant
- ❌ energy hungry
- ❌ unable to make sense of situations that deviate from their training data
- ❌ Moravec's paradox

Can we build more intelligent systems if we understand our own intelligence first?



What is cognitive science?

The study of mental process

- Perception
- Personality
- Language
- Attention
- Learning and memory

Cognition is the process by which the sensory input is transformed, reduced, elaborated, stored, recovered, and used *(Neisser, 1967)*

Work from Cognitive Psicology has informed and influenced virtually all other areas of Psychology and AI!

Example: Perception

Example of "top-down" control: our perception, even on a low level, is influenced by our previous knowledge and concepts

Example: Attention

Selective attention test

https://www.youtube.com/watch?v=vJG698U2Mvo

Gradual Change test

https://www.youtube.com/watch?v=1nL5ulsWMYc

Q: Is this a desirable feature for artificially intelligent systems?

Example: Personality

- **Example: Personality**
  - Personality (extroverted, polite, persistent, curious…) influences our thoughts and actions
  - Study: surprising links between human cognition and personality (millions of people across 50+ countries)

PNAS RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES

Check for updates

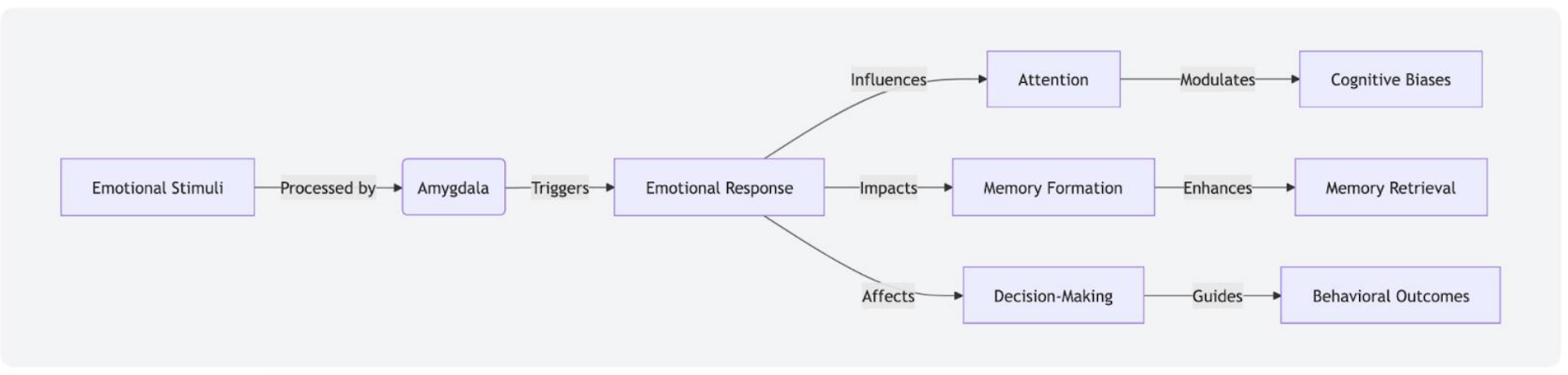
Meta-analytic relations between personality and cognitive ability

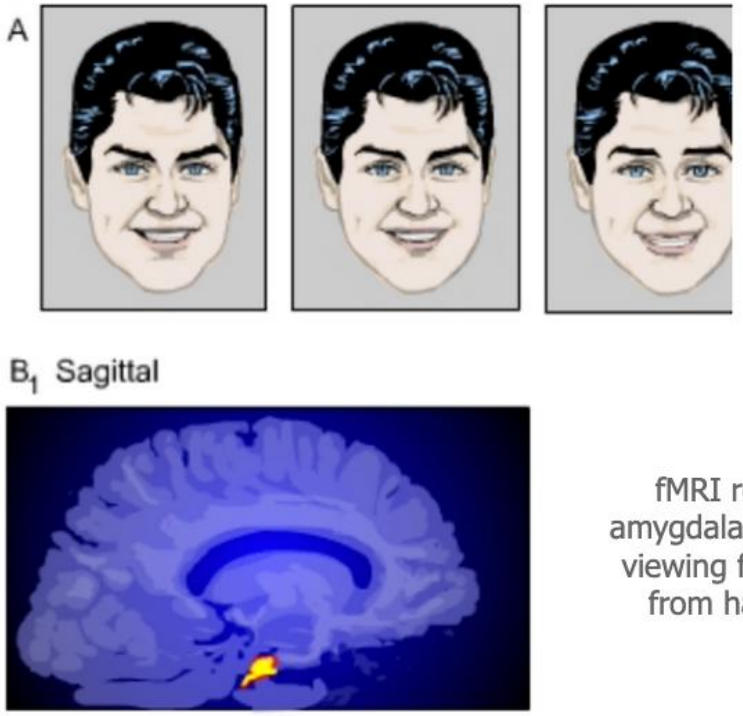
Kevin C. Stanek[a,1] and Deniz S. Ones[a,1]

Edited by Daniel J. Benjamin, University of California, Los Angeles, CA; received August 22, 2022; accepted March 24, 2023 by Editorial Board Member Michael S. Gazzaniga

  - E.g., those who were more compassionate tended to have better verbal and quantitative knowledge skills, suggesting a connection between personality traits and how we learn.

Example: Emotions

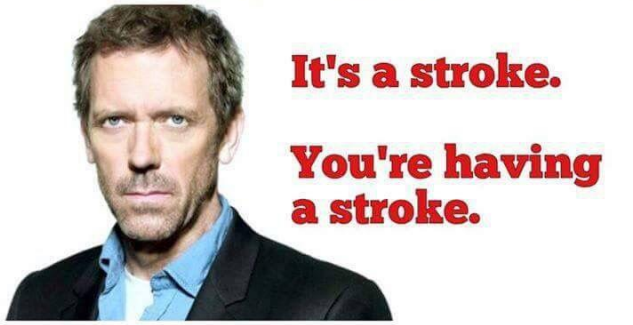




fMRI results showing amygdala activity in normal viewing facial expressions from happy to fearful.

https://nba.uth.tmc.edu/neuroscience/m/s4/chapter06.html

Example: Language



Garden-path sentences:

- The old man the boat.
- The complex houses married and single soldiers and their families.
- Fat people eat accumulates.

Q: Why do we have trouble reading sentences such as this one?

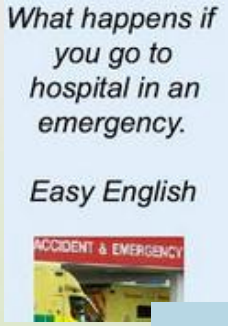
I.e., how does the **parsing** of a sentence work?

Your move

- Examples of real world applications –

Research into ***Attention***:

Research into ***Long term memory & Learning***:

Research into ***Language***:

Perception & Short term memory: Improvements in human-computer interaction